

# Neural Networks and Deep Learning

## Lecture 11: Explainable AI: SHAP, LIME, gradient methods and the stability of explanations





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## Learning objectives

- Explain why banks, insurers and regulators require explanations of a neural network, and classify explanation methods of explainable artificial intelligence (XAI).
- Compute Shapley additive explanations (SHAP) and local interpretable model-agnostic explanations (LIME) with full formulas, and read them for one customer and for a portfolio.
- Use gradient methods (saliency, integrated gradients, SmoothGrad) and check their axioms on a real fraud model.
- Test whether explanations agree, are faithful to the model and stay stable across seeds, samples and time.
- Write a counterfactual explanation that a rejected applicant can act on.

# Abbreviations

| | |
|---|---|
| **AI** | artificial intelligence |
| **AOPC** | area over the perturbation curve |
| **ASE** | Bucharest University of Economic Studies |
| **AUC** | area under the ROC curve |
| **EBA** | European Banking Authority |
| **EU** | European Union |
| **EUR** | euro |
| **GDPR** | General Data Protection Regulation |
| **GNN** | graph neural network |
| **HMEQ** | home-equity loan data set |
| **ICE** | individual conditional expectation |
| **IG** | integrated gradients |
| **IPE** | Institute for Economic Forecasting, Romanian Academy |
| **JS** | Jensen-Shannon divergence |
| **KL** | Kullback-Leibler divergence |
| **LIME** | local interpretable model-agnostic explanations |
| **MAD** | median absolute deviation |
| **MLP** | multilayer perceptron |
| **PCA** | principal component analysis |
| **PD** | probability of default |
| **PDP** | partial dependence plot |
| **PICBE** | Proceedings of the International Conference on Business Excellence |
| **PR-AUC** | area under the precision-recall curve |
| **ReLU** | rectified linear unit |
| **ROC** | receiver operating characteristic |
| **SHAP** | Shapley additive explanations |
| **SR** | Supervisory Guidance on Model Risk Management (Federal Reserve SR 11-7) |
| **TCAV** | testing with concept activation vectors |
| **ULB** | Université Libre de Bruxelles |
| **USD** | US dollar |
| **XAI** | explainable artificial intelligence |
| **XGBoost** | extreme gradient boosting |

# Real data in this lecture

| Case | Data | Models and split |
|---|---|---|
| Credit | Home-equity loan data set (HMEQ), 5 960 loans, 19.9 % defaults (Baesens et al., 2016) | multilayer perceptron (MLP), logistic regression, extreme gradient boosting (XGBoost); 60/20/20 stratified |
| Fraud | Credit-card transactions of the Université Libre de Bruxelles (ULB) data set, two days of September 2013; 281 + 211 frauds in the sample (Dal Pozzolo et al., 2015) | MLP and logistic regression; train on day 1, test on day 2 |
| Housing | California housing, 20 640 block groups of the 1990 census | MLP regression (partial dependence) |

- Every number is computed by analysis_L11.py
  - Seeds from 20261108
  - Explanations computed on an Apple M5 Max processor.
- Fraud variables V1–V28 are principal component analysis (PCA) components released by the data owner: explanations name components, not raw fields.

# Outline

- Reasons for explaining neural networks
- Shapley values and SHAP
- LIME, counterfactuals and global effects
- Gradient methods for neural networks
- Agreement, faithfulness and stability of explanations
- Explainability in model governance

# Regulators require explanations of automated decisions

- **European Union (EU) Artificial Intelligence (AI) Act** (Regulation 2024/1689): creditworthiness scoring of natural persons is high-risk (Annex III)
  - Transparency to deployers (art. 13) and a right to an explanation of individual decisions (art. 86).
- **General Data Protection Regulation (GDPR)**, art. 22 and recital 71: safeguards against decisions based solely on automated processing.
- **European Banking Authority (EBA)**, guidelines on loan origination and monitoring (2020): models used for credit decisions must be explainable.
- **Model risk**: the Federal Reserve Supervisory Guidance on Model Risk Management (SR 11-7) asks for conceptual soundness, independent validation and ongoing monitoring.
- **Business reasons**: finding leakage, trust of credit officers, explaining a rejection.
- **But**: an explanation is a model of the model, and it can be wrong.

- **In practice:** an explanation that changes when nothing important changed is a compliance risk, not a safeguard.

(Regulation (EU) 2024/1689; SR 11-7)

# A taxonomy of explanation methods

| Method | Scope | Needs | This lecture |
|---|---|---|---|
| Coefficients, scorecards | global, intrinsic | a linear model | logistic benchmark |
| SHAP | local $\rightarrow$ global | model calls (Kernel); trees (TreeSHAP) | credit MLP, XGBoost |
| LIME | local | model calls | same applicant |
| Saliency, integrated gradients (IG), SmoothGrad | local | gradients | fraud MLP |
| Partial dependence plot (PDP), individual conditional expectation (ICE) | global effect | model calls | housing MLP |
| Counterfactuals | local, actionable | gradients or search | rejected applicant |
| Permutation importance | global | labels + model calls | fraud MLP |

- Post-hoc methods explain a trained black box
  - Interpretable models are transparent by design (Rudin, 2019).

(Rudin, 2019)

# Credit case study: comparison of three scoring models

- **In one sentence:** the network ranks defaulters better than the logistic scorecard (higher area under the receiver operating characteristic (ROC) curve (AUC) and area under the precision-recall curve (PR-AUC)); can we explain it as well?

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \quad \mathbf{h}_2 = \mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)$$
$$f(\mathbf{x}) = \mathbf{w}_3^\top \mathbf{h}_2 + b_3, \qquad \mathrm{PD}(\mathbf{x}) = \sigma\big(f(\mathbf{x})\big)$$

| Model (test) | AUC | PR-AUC | Brier |
|---|---|---|---|
| Logistic | 0.909 | 0.785 | 0.080 |
| MLP, 10 seeds | 0.935 | 0.842 | 0.068 |
| XGBoost, 10 | 0.959 | 0.889 | 0.058 |

- Base rate (PR-AUC of a random score) 0.200.

- **where**
  - $\mathbf{x}$ = 18 standardised features, median-imputed, with a flag for a missing debt-to-income ratio (DEBTINC)
  - $f$ = log-odds of default
  - 64 and 32 rectified linear units (ReLU), dropout 0.1
  - $\mathrm{PD}$ = probability of default (PD).

- **In practice:** the logistic coefficients are the explanation
  - For the MLP the explanation must be computed.

(Baesens, Roesch & Scheule, 2016; own computation)

**Experiment yourself:** notebook §1

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# Shapley values: a fair split of the prediction among features

- **In one sentence:** each feature receives its average marginal contribution over all orders in which features could join the prediction.

$$\phi_j = \sum_{S \subseteq \{1,\dots,d\} \setminus \{j\}} \frac{|S|!\,(d - |S| - 1)!}{d!} \Big[ v(S \cup \{j\}) - v(S) \Big]$$

- **where**
  - $d$ = number of features
  - $v(S)$ = value of the coalition $S$ (what the model predicts when only the features in $S$ are known)
  - The weight is the share of orderings in which $j$ joins right after $S$.

- **The four axioms** (Shapley, 1953) that make $\phi$ unique
  - Efficiency: $\sum_j \phi_j = v(\{1..d\}) - v(\emptyset)$
  - Symmetry: equal contributions, equal shares
  - Dummy: a feature that never changes $v$ gets 0
  - Additivity: $\phi(v_1 + v_2) = \phi(v_1) + \phi(v_2)$
- **Cost**: $2^d$ coalitions
  - With 18 features, $2^{18}$ model values per applicant: we need approximations.

- **In practice:** efficiency is what a credit officer needs: the parts add up exactly to the score.

(Shapley, 1953)

# SHAP: the value of a coalition of features

- **In one sentence:** SHAP sets $v(S)$ = expected model output when the features outside $S$ are drawn from a background sample.

$$v(S) = \mathbb{E}_{\mathbf{x}_{\bar{S}} \sim \mathcal{B}}\left[f(\mathbf{x}_S, \mathbf{x}_{\bar{S}})\right]$$

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{d} \phi_j(\mathbf{x}), \qquad \phi_0 = \mathbb{E}_{\mathcal{B}}[f]$$

- **where**
  - $\mathcal{B} = 100$ training applicants (background)
  - Interventional expectation: features outside $S$ are replaced independently of $\mathbf{x}_S$
  - Explanations are on the log-odds scale, where effects add.

- **Additivity on our data**: largest error $|\phi_0 + \sum_j \phi_j - f(\mathbf{x})| = 2.9\text{e-}06$ over 400 test applicants.
- **Linear model**: $\phi_j = \beta_j(x_j - \mathbb{E}x_j)$ exactly (linear SHAP).
- **Off-manifold risk**: with correlated features, independent replacement creates unrealistic applicants.

- **In practice:** the background sample defines the "average applicant"
  - Document it like any model input.

(Lundberg & Lee, 2017)

**Experiment yourself:** notebook §2

# Kernel SHAP and TreeSHAP: how SHAP is computed

- **In one sentence:** Kernel SHAP fits a weighted linear regression on sampled coalitions; for trees the exact values come in polynomial time.

$$\min_{\phi} \sum_{\mathbf{z} \in \{0,1\}^d} \pi(\mathbf{z}) \left[ v(\mathbf{z}) - \phi_0 - \sum_j \phi_j z_j \right]^2$$

$$\pi(\mathbf{z}) = \frac{d-1}{\binom{d}{|\mathbf{z}|} \, |\mathbf{z}| \, (d - |\mathbf{z}|)}$$

- **where**
  - $\mathbf{z}$ = coalition indicator
  - The Shapley kernel $\pi$ makes the weighted least-squares solution equal the Shapley values (Appendix A)
  - We sample 600 coalitions per applicant.

- **Kernel SHAP**: model-agnostic
  - 16 s for 400 applicants.
- **TreeSHAP**: exact for tree ensembles such as XGBoost (Lundberg et al., 2020).
- **Deep networks**: Deep/Gradient SHAP combine Shapley values with gradients (Section 3).

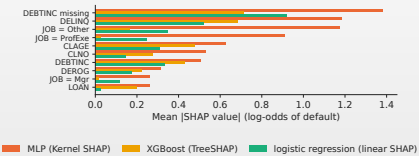
- **In practice:** Kernel SHAP has sampling noise: fix the seed and the number of samples in the model documentation.

(Lundberg & Lee, 2017; Lundberg et al., 2020)

**Experiment yourself:** notebook §2

# Global importance: the missing debt-to-income ratio dominates

- **In one sentence:** mean absolute SHAP value per feature, for the network and the two benchmarks.




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: 400 test applicants, 100 background applicants.

- Top for all three models: DEBTINC_missing
  - MLP: then DELINQ and JOB_Other
  - Spearman correlation of importances MLP–XGBoost 0.55, MLP–logistic 0.79.
- Debt-to-income ratio (DEBTINC) missing for 21.3 % of loans
  - Default rate 62.0 % when missing vs 8.6 % when reported.

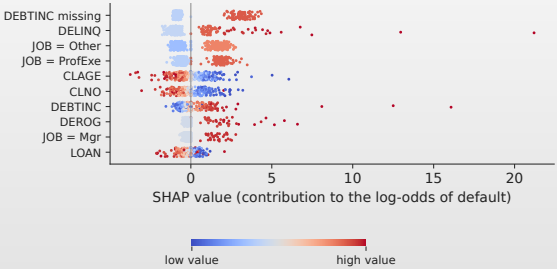
- **In practice:** an explanation can reveal that a model scores the paperwork rather than the borrower
  - Decide whether that is acceptable.

(own computation)

NN_L11_shap_global

# The beeswarm: direction and spread of every effect

- **In one sentence:** each dot is one applicant; colour = rank of the feature value; position = its SHAP value.




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: 400 test applicants; colour = rank of the feature value.

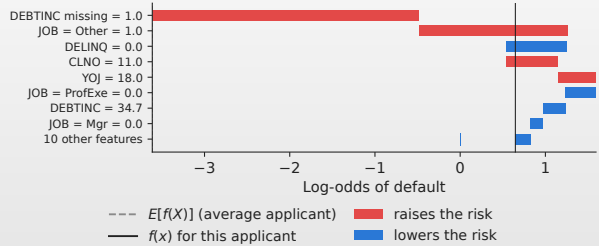
- Delinquent credit lines (DELINQ) and derogatory reports (DEROG): high values push the risk up, as credit theory expects.

(own computation)

NN_L11_shap_beeswarm

# Local explanation of a rejected applicant: SHAP waterfall

- **In one sentence:** from the average log-odds $\phi_0$ to this applicant's score, one feature at a time.




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: test applicant (row 2411); values in original units.

- Test applicant (row 2411): PD 65.6 %, log-odds $0.648 = -3.613 + 4.261$
  - Largest pushes: DEBTINC_missing ($+3.13$), JOB_Other ($+1.74$), DELINQ ($-0.71$).

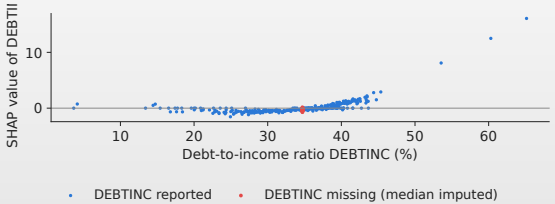
- **In practice:** the adverse-action notice lists the top reasons in this order, in words the customer understands.

(own computation)

NN_L11_shap_waterfall

# Dependence plot: the network learned the missing-value signal

- **In one sentence:** SHAP value of DEBTINC against its value; red points are imputed at the training median.




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: 400 test applicants, MLP.

- Imputed applicants sit at one DEBTINC value
  - Their risk is carried by the missing flag (mean SHAP +3.13), not by DEBTINC itself.

- **In practice:** imputation choices change explanations
  - Keep a missing-value flag so that the reason can be named.

(own computation) NN_L11_shap_dependence

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# LIME: a simple model that is faithful near one point

- **In one sentence:** perturb the applicant, weight the perturbations by closeness, fit a sparse linear model to the network's answers.

$$\xi(\mathbf{x}) = \arg\min_{g \in \mathcal{G}} \sum_{\mathbf{z}} \pi_{\mathbf{x}}(\mathbf{z}) \big[ f(\mathbf{z}) - g(\mathbf{z}) \big]^2 + \Omega(g)$$

$$\pi_{\mathbf{x}}(\mathbf{z}) = \exp\left( - \frac{\|\mathbf{z} - \mathbf{x}\|^2}{\sigma^2} \right)$$

- **where**
  - $\mathcal{G}$ = linear models $g(\mathbf{z}) = \beta_0 + \beta^\top \mathbf{z}$
  - $\Omega$ = complexity penalty
  - $\mathbf{z} \sim \mathcal{N}(\mathbf{x}, \mathbf{I})$ in standardised units
  - Kernel width $\sigma = 0.75\sqrt{d} = 3.18$ by default.

- **Local slope**, not a split of the prediction: the coefficients do not add up to $f(\mathbf{x})$.
- **Three choices** decide the answer: neighbourhood, kernel width, number of samples.
- **Our applicant**: LIME's top feature is DELINQ, SHAP's is DEBTINC_missing.

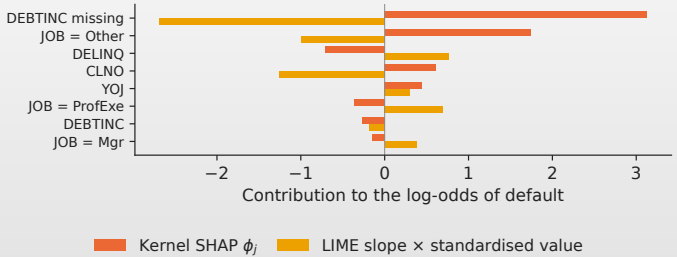
- **In practice:** LIME answers "what moves the score near this customer?", SHAP answers "why is this customer far from average?".

(Ribeiro, Singh & Guestrin, 2016)

**Experiment yourself:** notebook §3

# LIME and SHAP for the same applicant

- **In one sentence:** LIME slope times the standardised value, next to the SHAP contribution, for the eight largest SHAP features.




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: test applicant (row 2411); LIME with 5,000 samples.

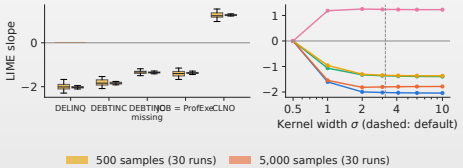
- Spearman correlation of the absolute contributions: 0.80
  - A binary flag is far from its local linear approximation.

(own computation)

NN_L11_lime_vs_shap

# Sensitivity of LIME to its settings

- **In one sentence:** left: 30 runs with different random seeds; right: the five slopes as the kernel width changes.




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: test applicant (row 2411); right panel: same five features, 5,000 samples.

- Same top feature: 73 % of runs with 500 samples, 100 % with 5,000
  - Jaccard overlap of the top-5 sets 0.72 vs 1.00
  - Coefficient of variation 10 % vs 2 %.

- **In practice:** fix and report the seed, sample size and kernel width of every LIME explanation that reaches a customer or an auditor.

(own computation)

NN_L11_lime_instability

# Counterfactual explanations: minimal changes that reverse the decision

- **In one sentence:** find the closest applicant, in actionable features only, whom the model would accept.

$$\mathbf{x}' = \arg\min_{\mathbf{x}'} \; \lambda \left( f(\mathbf{x}') - t \right)_+^2 + \sum_{j \in \mathcal{A}} \frac{|x'_j - x_j|}{\mathrm{MAD}_j}$$

- **where**
  - $t$ = target log-odds ($-0.5$, below the 0.5 PD threshold)
  - $\mathcal{A}$ = actionable features (LOAN, DEBTINC, NINQ, DELINQ, DEROG, CLNO, MORTDUE)
  - Median absolute deviation (MAD) scales the distance
  - Raw values kept $\geq 0$, counts rounded.

| Feature | now | needed |
|---|---|---|
| MORTDUE | 69 115.0 | 57 456.5 |
| DEBTINC | 34.7 | 33.4 |
| LOAN | 14 500.0 | 15 700.0 |

- PD 65.6 % → 39.2 % after these changes.
- Documentary counterfactual: reporting DEBTINC (at the median 34.7) gives PD 5.1 %.

- **In practice:** "bring the missing income document" is a better answer to the customer than any list of SHAP values.

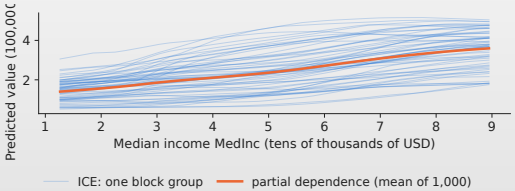
(Wachter, Mittelstadt & Russell, 2017)

**Experiment yourself:** notebook §4

# Global effects: partial dependence and ICE

- **In one sentence:** the PDP averages the prediction over the data while one feature is set to a grid value; ICE keeps one curve per observation.

$$\mathrm{PD}_j(z) = \frac{1}{n}\sum_{i=1}^{n} f(z, \mathbf{x}_{i,-j}), \qquad \mathrm{ICE}_{i,j}(z) = f(z, \mathbf{x}_{i,-j})$$




Source: StatLib, California housing, 1990 census (Pace and Barry, 1997). Data: MLP; 60 test block groups (ICE), 1,000 for the PDP.

- California housing MLP (test $R^2$ 0.819, linear 0.620): the predicted value rises from 1.40 to 3.60 (100,000 USD) over the income grid
  - Spread of the ICE curves at the top 0.95.

(Friedman, 2001; Goldstein et al., 2015)

NN_L11_pdp_ice

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# Fraud case study: a network trained on day 1 and explained on day 2

- **In one sentence:** gradients are free for a network: one backward pass gives the sensitivity of the fraud score to every input.

$$\text{saliency}_j(\mathbf{x}) = \left|\frac{\partial f(\mathbf{x})}{\partial x_j}\right|, \qquad \text{G}\times\text{I}_j(\mathbf{x}) = (x_j - x'_j)\,\frac{\partial f(\mathbf{x})}{\partial x_j}$$

- **where**
  - $f$ = fraud log-odds of an MLP on V1–V28 and log amount
  - $\mathbf{x}'$ = baseline = mean legitimate transaction of day 1
  - Class weight $\sqrt{\text{imbalance}}$, early stopping on weighted validation PR-AUC.

- **Test, day 2** (full-data weights): MLP PR-AUC 0.815, logistic 0.742
  - AUC 0.961 vs 0.964.
- **Saturation**: in a flat region of $f$ the gradient is near zero even for a decisive feature.
- **ReLU networks**: the gradient is piecewise constant, so saliency jumps between linear regions.

- **In practice:** a fraud analyst gets a ranked list of the inputs that drove the alert, in milliseconds per transaction.

(Simonyan, Vedaldi & Zisserman, 2014; Dal Pozzolo et al., 2015)

**Experiment yourself:** notebook §5

# Integrated gradients and the completeness of attributions

- **In one sentence:** integrate the gradient along the straight path from a baseline to the input; the attributions sum to $f(\mathbf{x}) - f(\mathbf{x}')$.

$$\mathrm{IG}_j(\mathbf{x}) = (x_j - x'_j) \int_0^1 \frac{\partial f\left(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')\right)}{\partial x_j}\, d\alpha$$
$$\approx (x_j - x'_j) \frac{1}{m} \sum_{k=1}^{m} \frac{\partial f\left(\mathbf{x}' + \frac{k-1/2}{m}(\mathbf{x} - \mathbf{x}')\right)}{\partial x_j}$$

- **where**
  - $m$ = Riemann steps (midpoint rule, 64 here)
  - Completeness $\sum_j \mathrm{IG}_j = f(\mathbf{x}) - f(\mathbf{x}')$ follows from the fundamental theorem of calculus for paths.

- **Axioms** (Sundararajan et al., 2017)
  - Completeness; sensitivity: a feature that changes $f$ gets non-zero credit
  - Implementation invariance: equal functions, equal attributions
- **Link to Shapley**: IG is the Aumann-Shapley value of the continuous game along the path.
- **Baseline matters**: "mean legitimate transaction" answers "why is this not a normal payment?".

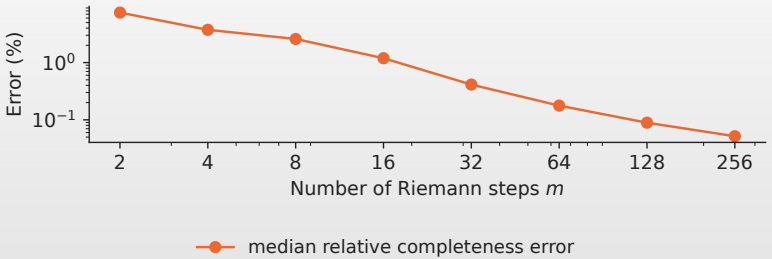
- **In practice:** write the baseline into the model documentation: a different baseline is a different question.

(Sundararajan, Taly & Yan, 2017)

**Experiment yourself:** notebook §5

# Completeness holds once the integral is computed accurately

- **In one sentence:** median of $|\sum_j \mathrm{IG}_j - (f(\mathbf{x}) - f(\mathbf{x}'))|/|f(\mathbf{x}) - f(\mathbf{x}')|$ over the frauds of day 2.




Source: ULB Machine Learning Group and Worldline, credit-card transactions (OpenML 1597). Data: 211 frauds of day 2; baseline = mean legitimate transaction of day 1.

- 3.7 % error with 4 steps, 0.18 % with 64, over 211 frauds
  - The error falls roughly as $1/m^2$, as for the midpoint rule.

(own computation)

NN_L11_ig_completeness

# SmoothGrad and concept-level explanations

- **In one sentence:** average gradients over noisy copies of the input to remove ReLU jitter; or ask whether a human concept moves the output.

$$\mathrm{SG}_j(\mathbf{x}) = \frac{1}{n}\sum_{k=1}^{n} \frac{\partial f(\mathbf{x}+\varepsilon_k)}{\partial x_j}, \quad \varepsilon_k \sim \mathcal{N}(\mathbf{0}, s^2\mathbf{I})$$

$$S_C(\mathbf{x}) = \nabla h(\mathbf{x})^\top \mathbf{v}_C$$

- **where**
  - $n = 50$, $s = 0.15$ standard deviations
  - Testing with concept activation vectors (TCAV): $\mathbf{v}_C =$ normal of a linear classifier separating examples of concept $C$ in a hidden layer $h$
  - The TCAV score is the share of inputs with $S_C > 0$.

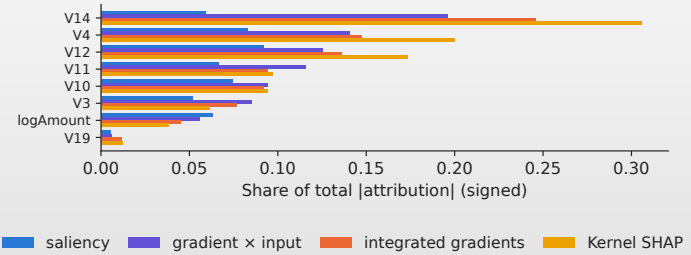
- **SmoothGrad** sharpens image saliency maps and stabilises tabular gradients (next section).
- **TCAV** speaks the language of the business: "night-time transaction", "new merchant" — if such concepts are labelled.
- **Our data**: V1–V28 are anonymised components, so concept tests are not possible here.

- **In practice:** concept explanations need concepts labelled when the data are collected.

(Smilkov et al., 2017; Kim et al., 2018)

# Four explanation methods for one fraudulent transaction

- **In one sentence:** signed attributions of one day-2 fraud (median fraud score), as shares of the total absolute attribution.




Source: ULB Machine Learning Group and Worldline, credit-card transactions (OpenML 1597). Data: one fraud of day 2 (median fraud score).

- Top feature: Kernel SHAP V14, integrated gradients V14, saliency V12
  - Fraud score 99.9 %: saliency is blind to the size of the deviation from the baseline.

(own computation)

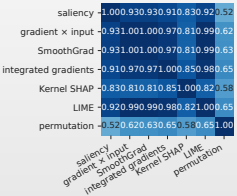
NN_L11_fraud_local

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# Agreement between explanation methods

- **In one sentence:** Spearman correlation of the global importances (mean |attribution|) of seven methods on the same frauds.




Source: ULB Machine Learning Group and Worldline, credit-card transactions (OpenML 1597). Data: Spearman correlation of global importances; 211 frauds of day 2.

- IG–SHAP 0.85, saliency–SHAP 0.83, LIME–SHAP 0.82, permutation–SHAP 0.58
  - All methods put V12 first.

- **In practice:** disagreement is information: permutation importance measures what the model needs, attributions what it uses.

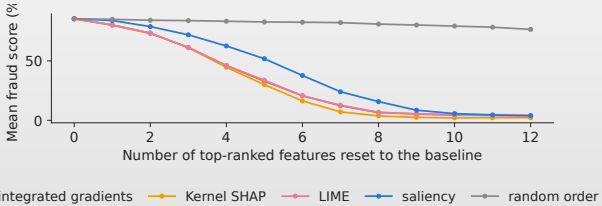
(Breiman, 2001; own computation)

NN_L11_method_agreement

# Faithfulness of explanations: deletion test

- **In one sentence:** reset the top-$k$ features of each fraud to the baseline; a faithful explanation makes the score fall fast.

$$\mathrm{AOPC} = \frac{1}{K}\sum_{k=1}^{K}\left[f(\mathbf{x}) - f(\mathbf{x}^{(k)})\right]$$




Source: ULB Machine Learning Group and Worldline, credit-card transactions (OpenML 1597). Data: 211 frauds of day 2.

- Area over the perturbation curve (AOPC) in points of fraud score: IG 56.2, SHAP 58.4, LIME 56.1, saliency 48.0, random order 3.9.

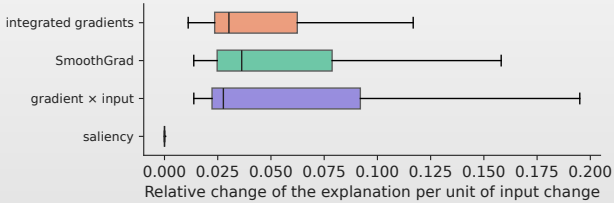
(Samek et al., 2017)

NN_L11_deletion_curve

# Local robustness of explanations to small input changes

- **In one sentence:** relative change of the attribution vector per unit of a small perturbation (Lipschitz-type ratio).

$$\hat{L}(\mathbf{x}) = \frac{\|\mathbf{a}(\mathbf{x}+\varepsilon) - \mathbf{a}(\mathbf{x})\| / \|\mathbf{a}(\mathbf{x})\|}{\|\varepsilon\|}, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, 0.05^2\mathbf{I})$$




Source: ULB Machine Learning Group and Worldline, credit-card transactions (OpenML 1597). Data: 60 frauds of day 2 x 10 perturbations (sigma = 0.05 standard deviations).

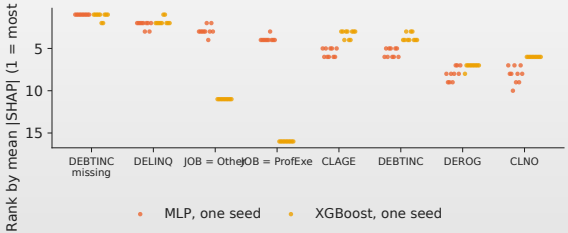
- Medians: saliency 0.000, gradient $\times$ input 0.028, IG 0.030, SmoothGrad 0.036
  - Within a linear region of a ReLU network the gradient does not move.

(Alvarez-Melis & Jaakkola, 2018)

NN_L11_local_robustness

# The Rashomon effect: equally accurate models with different explanations

- **In one sentence:** rank of each feature by mean |SHAP| for 10 retrained networks and 10 boosted ensembles (different seeds only).




Source: Baesens, Roesch and Scheule (2016), HMEQ data. Data: 10 training seeds; 200 test applicants.

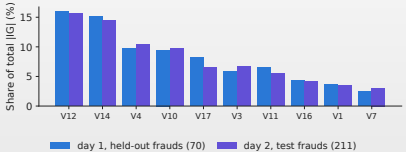
- MLP test AUC between 0.929 and 0.941
  - Mean pairwise Kendall $\tau$ (rank agreement of importances, 1 = identical order): MLP 0.83, XGBoost 0.97
  - Same top feature for one applicant in 78 % (MLP) and 83 % (XGBoost) of seed pairs.

(own computation)

NN_L11_seed_stability

# Explanations over time: day 1 against day 2

- **In one sentence:** share of the total |IG| by feature, for held-out frauds of the training day and for the frauds of the next day.




Source: ULB Machine Learning Group and Worldline, credit-card transactions (OpenML 1597). Data: model trained on day 1; integrated gradients.

- Spearman 0.98, Jensen-Shannon (JS) divergence (built on the Kullback-Leibler (KL) divergence) 0.003, 4 of the top 5 shared
  - One day later the explanation is stable here.

- **In practice:** monitor the distribution of attributions monthly, next to the score distribution
  - A drift in reasons can precede a drift in accuracy.

(own computation) NN_L11_time_shift

## Stability scorecard of this lecture

| Source of change | Measure | Result |
|---|---|---|
| Explainer randomness (LIME) | same top feature over 30 runs | 73 % (500), 100 % (5,000 samples) |
| Training seed (Rashomon) | Kendall $\tau$ of global importances | MLP 0.83, XGBoost 0.97 |
| Training seed, one applicant | same top feature | MLP 78 %, XGBoost 83 % |
| Method | Spearman of importances | 0.52 – 1.00 |
| Input noise | median relative change | IG 0.030, SmoothGrad 0.036 |
| Time (day 1 $\rightarrow$ day 2) | Spearman, JS divergence | 0.98; 0.003 |

⊡ **In practice:** report every explanation with its stability: which seed, which sample, which period, and how much it moves.

(own computation)

# Outline

- Reasons for explaining neural networks
- Shapley values and SHAP
- LIME, counterfactuals and global effects
- Gradient methods for neural networks
- Agreement, faithfulness and stability of explanations
- Explainability in model governance

# An explainability audit checklist

- **Purpose**: who reads the explanation (customer, analyst, validator, regulator) and what decision it supports.
- **Settings**: background sample, baseline, seed, number of samples, kernel width — versioned with the model.
- **Checks**: additivity/completeness, faithfulness (deletion), stability (seeds, time), agreement of two methods.
- **Fairness**: attributions of protected attributes and of their proxies (age, sex, postcode)
  - An explanation shows use, not causation.
- **Customer output**: few reasons, actionable counterfactuals, no raw component names (V14 means nothing to a customer).
- **Alternative**: if an interpretable model is almost as accurate (here logistic AUC 0.909 vs MLP 0.935), consider using it.

**In practice:** the validation report should contain the explanations and their stability tests, not only the accuracy.

(SR 11-7; Rudin, 2019)

## Quiz 1: adding up SHAP values

- For our applicant the base value is $-3.613$ and the SHAP values sum to 4.261.
  - What is the model's log-odds of default, and why can you be sure?

**Answer**

- 0.648: by the efficiency axiom $f(\mathbf{x}) = \phi_0 + \sum_j \phi_j$
  - The PD is $\sigma(0.648) = 65.6\,\%$.

## Quiz 2: an unstable LIME

- Two analysts explain the same rejected applicant with LIME and report different top reasons.
  - Is the model inconsistent?

**Answer**

- Not necessarily: LIME samples perturbations
  - With 500 samples the top feature matched in 73 % of our runs
  - Fix the seed, raise the samples, check against SHAP.

## Quiz 3: choosing an IG baseline

- A colleague uses the all-zero vector as the integrated-gradients baseline for standardised fraud features.
  - What question does that explanation answer?

**Answer**

- "Why does this transaction differ from the average transaction (frauds included)?"
  - Our baseline, the mean legitimate transaction, asks why it is not a normal payment.

## Wrap-up

- **SHAP**: Shapley values with an exact additivity check
  - The missing debt-to-income ratio drives the credit network.
- **LIME**: a local slope that depends on seed, sample size and kernel width
  - Useful, but report its settings.
- **Gradients**: integrated gradients satisfy completeness (0.18 % error at 64 steps) and are as faithful as SHAP at a fraction of the cost.
- **Stability**: retraining with another seed reorders features (Kendall $\tau$ 0.83 for the MLP)
  - Explanations must be tested like models.
- **Next**: Lecture 12, large language models, where explanations become text.

# Research from our group: XAI in fraud and energy markets

- **SHAP stability in graph-augmented fraud detection**: on the IEEE-CIS fraud data, SHAP contributions of an XGBoost model with graph neural network (GNN) embeddings are compared under random and forward-time validation, with a Gaussian control of the same dimension and covariance
  - Whether the graph model looks more stable depends on how the change is measured, a careful protocol for explanation stability (Diaconu & Pele, 2026, submitted to Expert Systems with Applications; code: QuantLet GNN_XAI_Stability).
- **Explainable early warning of electricity price spikes** in Romania (Andrei, Pele, Cramer, Amza & Diaconu, 2026, Proceedings of the International Conference on Business Excellence (PICBE)) and **SHAP-driven anomaly detection** of electricity prices (Andrei & Pele, 2025, Preprints).
- **Multi-horizon explainability** of price forecasts: which drivers matter at which horizon in Romania's liberalised market (Andrei, Pele & Cramer, 2026, PICBE).
- Findings as stated in the papers; numbers on the other slides come from our own data.

(Bucharest University of Economic Studies (ASE); Romanian Academy, Institute for Economic Forecasting (IPE))

## Further reading and resources

- **C. Molnar, Interpretable Machine Learning**, free online: christophm.github.io/interpretable-ml-book.
- **SHAP library** and examples: shap.readthedocs.io.
- **Rudin (2019)**: the case for interpretable models in high-stakes decisions.
- **This course**: code, data, notebooks — github.com/danpele/NEURAL_BIZ.
- **Experiment yourself**: notebooks/L11_notebook.ipynb
  - The seminar explains a German credit model, with a fairness check.
- **Replication of our stability study**: github.com/QuantLet/GNN_XAI_Stability.

# References (1/3)


- Alvarez-Melis, D. & Jaakkola, T. S. (2018). On the robustness of interpretability methods. ICML Workshop on Human Interpretability in Machine Learning. arXiv:1806.08049
- Andrei, A.-V. & Pele, D. T. (2025). Explainable AI for electricity price anomaly detection: A SHAP-driven approach in Romania's energy market. *Preprints*. doi:10.20944/preprints202508.0410.v1
- Andrei, A.-V., Pele, D.-T. & Cramer, A.-A. (2026). Multi-horizon explainability in energy price forecasting: Disentangling market dynamics and renewable intermittency in Romania's liberalized electricity market. *Proceedings of the International Conference on Business Excellence*, 20(1), 104–117. doi:10.2478/picbe-2026-0014
- Andrei, A.-V., Pele, D.-T., Cramer, A.-A., Amza, A. & Diaconu, D. (2026). Explainable AI-based early warning system for electricity price spikes: Evidence from the Romanian market. *Proceedings of the International Conference on Business Excellence*, 20(1), 4820–4836. doi:10.2478/picbe-2026-0352
- Baesens, B., Roesch, D. & Scheule, H. (2016). *Credit Risk Analytics: Measurement Techniques, Applications, and Examples in SAS*. Wiley.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32. doi:10.1023/A:1010933404324
- Dal Pozzolo, A., Caelen, O., Johnson, R. A. & Bontempi, G. (2015). Calibrating probability with undersampling for unbalanced classification. IEEE SSCI, 159–166. doi:10.1109/SSCI.2015.33
- Diaconu, D. & Pele, D. T. (2026). SHAP stability across random and forward-time validation in graph-augmented fraud detection. Submitted to *Expert Systems with Applications*. Code: github.com/QuantLet/GNN_XAI_Stability

# References (2/3)


- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232. doi:10.1214/aos/1013203451
- Goldstein, A., Kapelner, A., Bleich, J. & Pitkin, E. (2015). Peeking inside the black box: Visualizing statistical learning with plots of individual conditional expectation. *Journal of Computational and Graphical Statistics*, 24(1), 44–65. doi:10.1080/10618600.2014.907095
- Kim, B. et al. (2018). Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). ICML. arXiv:1711.11279
- Lundberg, S. M. & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. arXiv:1705.07874
- Lundberg, S. M. et al. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2, 56–67. doi:10.1038/s42256-019-0138-9
- Ribeiro, M. T., Singh, S. & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. KDD, 1135–1144. doi:10.1145/2939672.2939778
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1, 206–215. doi:10.1038/s42256-019-0048-x
- Samek, W., Binder, A., Montavon, G., Lapuschkin, S. & Müller, K.-R. (2017). Evaluating the visualization of what a deep neural network has learned. *IEEE Transactions on Neural Networks and Learning Systems*, 28(11), 2660–2673. doi:10.1109/TNNLS.2016.2599820

# References (3/3)


- Shapley, L. S. (1953). A value for n-person games. In *Contributions to the Theory of Games II*, 307–317. Princeton University Press.
- Simonyan, K., Vedaldi, A. & Zisserman, A. (2014). Deep inside convolutional networks: Visualising image classification models and saliency maps. ICLR Workshop. arXiv:1312.6034
- Smilkov, D., Thorat, N., Kim, B., Viégas, F. & Wattenberg, M. (2017). SmoothGrad: Removing noise by adding noise. arXiv:1706.03825
- Sundararajan, M., Taly, A. & Yan, Q. (2017). Axiomatic attribution for deep networks. ICML, 3319–3328. arXiv:1703.01365
- Wachter, S., Mittelstadt, B. & Russell, C. (2017). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2), 841–887. doi:10.2139/ssrn.3063289
- Regulation (EU) 2024/1689 (Artificial Intelligence Act), *Official Journal of the European Union*, L, 12 July 2024. data.europa.eu/eli/reg/2024/1689/oj
- Board of Governors of the Federal Reserve System (2011). SR 11-7: Supervisory guidance on model risk management. federalreserve.gov

# Appendix A: recovery of Shapley values by the Shapley kernel

- Restrict $g(\mathbf{z}) = \phi_0 + \sum_j \phi_j z_j$ to satisfy $g(\mathbf{0}) = v(\emptyset)$ and $g(\mathbf{1}) = v(\{1..d\})$ (infinite weight on the empty and full coalitions).
- The weighted least-squares normal equations with $\pi(\mathbf{z}) \propto (d-1)/[\binom{d}{|\mathbf{z}|}|\mathbf{z}|(d-|\mathbf{z}|)]$ have the unique solution $\phi_j =$ Shapley value (Lundberg & Lee, 2017, Theorem 2).
- Kernel SHAP samples coalitions from $\pi$ (pairing $\mathbf{z}$ with $\mathbf{1}-\mathbf{z}$) and solves the constrained regression
  - The error falls as $1/\sqrt{\text{samples}}$.
- For a linear $f$ and independent features, $v(S) = \beta_0 + \sum_{j\in S}\beta_j x_j + \sum_{j\notin S}\beta_j \mathbb{E}x_j$, so $\phi_j = \beta_j(x_j - \mathbb{E}x_j)$.

(Lundberg & Lee, 2017)

# Appendix B: integrated gradients as a path method

- For a path $\gamma(\alpha)$ from $\mathbf{x}'$ to $\mathbf{x}$: $f(\mathbf{x}) - f(\mathbf{x}') = \int_0^1 \nabla f(\gamma(\alpha))^\top \gamma'(\alpha)\, d\alpha = \sum_j \int_0^1 \partial_j f(\gamma)\, \gamma_j'\, d\alpha$.
- The straight line $\gamma(\alpha) = \mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')$ gives $\gamma_j' = x_j - x_j'$: each term is $\mathrm{IG}_j$, hence completeness.
- Among path methods, the straight line is the only one that is symmetric in features (Sundararajan et al., 2017).
- Midpoint Riemann sum error is $O(m^{-2})$ for smooth $f$
  - ReLU kinks make it slower, as our completeness chart shows at small $m$.

(Sundararajan, Taly & Yan, 2017)